William C. Choi-Kim

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RESEARCH EXPERIENCE

UT Health / University of Indiana **Jan. 2024 – Present**
Research Intern *Hybrid*

- **Sole research intern** working on Conjunctive Bayesian Networks (CBNs) under **Dr. Rzgar Hosseini**
  - Developed a R package (CBN2Path) to implement CT-CBN, H-CBN, and B-CBN in the language (https://bioconductor.org/packages/CBN2Path/)
  - First author research manuscript published in F1000 (https://doi.org/10.12688/f1000research.168810.1)
  - Used R's C interface and package development pipeline

Baylor College of Medicine **Jan 2025 - Present**
Research intern *Remote*

- Working with research data in the R programming language to do statistical analysis on nutritional and behavioural science regarding obesity

Wolfram Research **Aug. 2024 – Present**
Student Ambassador *Remote*

- Publish **monthly research essays** on the Wolfram Community platform
  - **Five *Staff Picks*** as a Student Ambassador
  - **Spoke** at the Wolfram Technology Conference 2025

School of Medicine Pathways Program at Johns Hopkins **2023 – 2024**
Research Intern *Hybrid*

- Completed a summer internship in oncology research at Johns Hopkins.
  - Completed a research project for the **Sidney Kimmel 31st Trainee Day**

LEADERSHIP EXPERIENCE

Hwalgi **May. 2024 – Present**
Founder

- Founded and run an international nonprofit that provides a platform for student essay writers to **share their opinions on political and social justice** with the world
  - Essays published from **the Philippines, Indonesia, India, the United States, and other nations**

Empty Streets **Sep. 2024 – Present**
Chief Technology Officer

- Built and maintain a website for **the world's largest youth-led anti-human-trafficking organization**

Crafts and Laughs **Jul. 2024 – Present**
Co-Founder and Chief Technology Officer

- Built and maintain a website for a nonprofit creating toys for children in hospitals

Competitive Math Club **2023 - Present**
Founder and President

- Founded and lead a competitive math club that participates in the AMC and AIME competitions yearly for the first time at the school, grew to over a dozen competition members

Computer Science Club **2025 - Present**
President and Competitive Team Captain

- Planned, presented, and ran meetings for general members of large Computer Science Club (~30 members)
- Led and competed with varsity competitive programming team

National Honor Society **2025 - Present**
Internal Vice President

- Managed volunteer registration, website, internal communications, and tracking hours for NHS (200+ members)

Boy Scouts **2023 - 2024**
Senior Patrol Leader

- Planned, organized, and led meetings, campouts, recruiting events, and leadership meetings for Troop 901 (~48 members)

VOLUNTEERING

Reach Unlimited **2023 - Present**

Weekly

- Work with developmentally disabled group-home members weekly
- Help with setting up special events and festivals

Cy-Fair Helping Hands Community Garden **2024 – Present**

Weekly

- Maintain community garden for disadvantaged communities
- Completed Boy Scouts Eagle Project by building a concrete soil bin out of brick and mortar

Boy Scouts **2019 – 2024**

Service Projects

- Led and participated in Boy Scout service projects
- Organized and fundraised capstone project, led a dozen volunteers

National Honor Society **2024 - Present**

School Club

- Completed service hours as a member and later officer of Cy-Fair High School National Honor Society

Science National Honor Society **2024 - Present**

School Club

- Completed service hours as a member of Cy-Fair High School Science National Honor Society

SKILLS

- Computational Research (Computational Biology) - see *Research Experience*
- Programming (**https://github.com/rockwillck**)
  - R (intermediate)
  - JavaScript (advanced)
  - WolframLanguage (advanced)
  - Python (intermediate)
- Critical Essay Writing

WORK EXPERIENCE

Tutoring **2024 - Present**

- Tutored students in math classes, SAT (English and Math), and AP exam prep

Teaching Assistant at Wolfram Summer Research Program **Summer 2025**

- Helped summer program students with computational research at WSRP'25

Mentor at Wolfram Emerging Leaders Program **2025 - 2026**

- Working as a mentor at the Wolfram Emerging Leaders Program, guiding students through a yearlong research project

EDUCATION

Cy-Fair High School **2022 - 2026**

Expected Graduation: 2026 *Cypress, TX*

- Class Rank: 1/774
- SAT: 1590/1600
- PSAT: 1520/1520
- AP: Biology (5), Calculus BC (5), English Language (5), Physics C Mechanics (5), Physics C E&M (5), United States History (5), Computer Science A (5), Physics 1 (5), Precalculus (5), Spanish Language and Culture (5), Statistics (5), World History: Modern (5)
  - Currently taking: English Literature, Chemistry, Psychology, U.S. Government, Macroeconomics

PUBLICATIONS

- Choi-Kim W and Hosseini SR. CBN2Path: an R/Bioconductor package for the analysis of cancer progression pathways using Conjunctive Bayesian Networks. F1000Research 2025, 14:834 (https://doi.org/10.12688/f1000research.168810.1)
- Choi-Kim, W., Raja, S., & Srinivasan, A. (2025, February 4). [WELP24] Established and novel methods for protein analysis in Wolfram Language - Online Technical Discussion Groups—Wolfram Community. Wolfram Community: Connect with other Users of Mathematica and Wolfram Technologies. https://community.wolfram.com/groups/-/m/t/3379964
- Choi-Kim, W. (2024, July 11). [WSRP24] Exploring isomerism in coordination complexes - Online Technical Discussion Groups—Wolfram Community. Wolfram Community: Connect with other Users of Mathematica and Wolfram Technologies. https://community.wolfram.com/groups/-/m/t/3213707?p_p_auth=hL24ZW2z

AWARDS (full list available at: https://www.williamck.com)

National Awards

- Scholastic National Gold Medal for Critical Essay [2022]
- AIME Qualification [2022, 2023, 2024, 2025]

- AMC 10 Distinction Award [2022, 2023, 2024]
- Eagle Scout [2024]
- National Merit Semifinalist [2025]

State Awards

- UIL Ready Writing State 4th Place [2025]

Regional Awards

- UIL Ready Writing Regional Champion [2024, 2025]
- Scholastic Regional Gold Key for Critical Essay [2022, 2023, 2024, 2025]
- Scholastic Regional Gold Key for Short Story [2022, 2023, 2025]
- HP Codewars 5th Place [2025]